Yeyuan Chen

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Experience

- **Bytedance SEED Intern**
  AI for Math — *Sept. 2026 –*
- **Microsoft Research Intern** — Redmond, US
  Design maximally recoverable local reconstruction codes in algorithm group — *May. 2026 – Aug. 2026*
- **Sensetime NLP Intern** — Beijing, China
  Large Language Model Pretraining (GPT3, T5) — *June. 2021 – Mar. 2022*

Education

- **University of Michigan** — Ann Arbor, US
  PhD in Computer Science and Engineering — *Aug. 2023 – Present*
- **University of Oxford** — Oxfordshire, UK
  Undergraduate Visiting Student in Computer Science — *Oct. 2021 – June. 2022*
- **Xi'an Jiaotong University** — Xi'an, China
  Bachelor of engineering in Computer Science, Honor class of CS; GPA: 4.03/4.3 — *Sept. 2019 – July. 2023*
- **Xi'an Jiaotong University** — Xi'an, China
  Young Gifted Class preparatory course — *Sept. 2017 – June. 2019*

Research Interests

I am broadly interested in theoretical computer science, in particular coding theory, computational algebra, spectral graph theory and combinatorics.

Publications/Manuscripts

For those papers in theoretical computer science, authors are usually sorted in alphabetical order. STOC/SODA are flagship conferences in theoretical computer science. NeurIPS and AAAI are leading conferences in Machine Learning.

- **Algorithmic List Decoding of Reed–Solomon Codes up to Capacity**
  Joshua Brakensiek, Yeyuan Chen, Aaron Putterman, Zihan Zhang, Kai Zhe Zheng
  - **Preprint**
- **Optimal Sparsifiers for Minkowski Sums and Sums of Seminorms**
  Arpon Basu, Joshua Brakensiek, Yeyuan Chen, Aaron Putterman, Victor Reis, Zihan Zhang
  - **Preprint**
- **Bounds and Limitations on Codes Achieving List Recovery Capacity**
  Joshua Brakensiek, Yeyuan Chen, Aaron Putterman, Zihan Zhang
  - **SODA 2027**
- **Unique Decoding of Reed-Solomon and Related Codes for Semi-Adversarial Errors**
  Joshua Brakensiek, Yeyuan Chen, Manik Dhar, Zihan Zhang
  - **ICALP 2026**
- **Combinatorial Bounds for List Recovery via Discrete Brascamp–Lieb Inequalities**
  Joshua Brakensiek, Yeyuan Chen, Manik Dhar, Zihan Zhang
  - **STOC 2026**
- **From Random to Explicit via Subspace Designs With Applications to Local Properties and Matroids**
  Joshua Brakensiek, Yeyuan Chen, Manik Dhar, Zihan Zhang
  - **STOC 2026**

- **Optimal Erasure Codes and Codes on Graphs**
  Yeyuan Chen, Mahdi Cheraghchi, Nikhil Shagrithaya
  - Preprint
- **Explicit Folded Reed-Solomon and Multiplicity Codes Achieve Relaxed Generalized Singleton Bounds**
  Yeyuan Chen, Zihan Zhang
  - **STOC 2025 Best Student Paper. Invited to SICOMP special issue**
- **The Gap Between Greedy Algorithm and Minimum Multiplicative Spanner**
  Yeyuan Chen
  - Preprint
- **Unique-neighbor Expanders with Better Expansion for Polynomial-sized Sets**
  Yeyuan Chen
  - **SODA 2025**
- **Calibrate and Boost Logical Expressiveness of GNN Over Multi-Relational and Temporal Graphs**
  Yeyuan Chen, Dingmin Wang
  - **NeurIPS 2023**
- **Range Avoidance, Remote Point, and Hard Partial Truth Table via Satisfying-Pairs Algorithms**
  Yeyuan Chen, Yizhi Huang, Jiatu Li, Hanlin Ren
  - **STOC 2023**
- **Efficient Embeddings of Logical Variables for Query Answering over Incomplete Knowledge Graphs**
  Dingmin Wang, Yeyuan Chen, Bernardo Cuenca Grau
  - **AAAI 2023**

Selected Honors and Awards

- **STOC 2025 Danny Lewin Best Student Paper Award**
- **Rackham Graduate Student Research Grant Award**
- **Jane Street Fellowship 2026 Finalist**
- **NeurIPS 2023 Scholar Award**
- **U Michigan CSE Honors Competition** 2nd Place 2024
- **International Collegiate Programming Contest**
  - **Bronze Medal** (13th place, 16th team) in 2022-2023 47th **World Finals** in Luxor
  - *1st place* in East Asia Regional Contest (Shanghai site 2020) (1/676) **Advances to World Finals 2021**
  - *2nd place* in East Asia Regional Contest (Hongkong site 2022) (2/115) **Advances to World Finals 2023**
  - *2nd place* in North American East Central Regional Contest (3rd place in the whole East) 2023
  - *4th place* in East Asia Regional Contest (Nanjing site 2022) (4/417)
  - *5th place* in Northwestern Europe Finals 2021 (5/129) (https://2021.nwerc.eu/standings/)
  - *15th place* in North American Finals 2024 **Advances to World Finals 2024** in Astana
- **China Collegiate Programming Contest**
  - *2nd place* in Qinhuangdao site 2020 (2/243), *3rd place* in Harbin site 2019 (3/272)
  - *4th place* in China Finals 2021 (4/115), *4th place* in Weihai site 2020 (4/267)
- **CCF Collegiate Computer Systems & Programming Contest** Gold Medal 2021
- **Scholarships:** Jieshan Scholarship 2021, XJTU First Class Scholarship 2021

Talks

- **U Michigan CSE Honors Competition 2024, Yao Seminar:** Explicit Folded Reed-Solomon and Multiplicity Codes Achieve Relaxed Generalized Singleton Bounds
- **SODA 2025:** Unique-neighbor Expanders with Better Expansion for Polynomial-sized Sets
- **STOC 2026:** From Random to Explicit via Subspace Designs With Applications to Local Properties and Matroids

Teaching

- **Graduate Student Instructor:** EECS 376 Foundations of Computer Science, Winter2025 Michigan

Service

- **Reviewer:** Multiple times for conferences ISIT, ICALP, SODA, STOC, CCC, FOCS, MFCS, RANDOM, ITW, Eurocrypt and journals IEEE Transactions on Communications, ACM Transactions on Algorithms, SIAM Journal on Computing.